

UNIT I: Introduction to Data Science

Data Science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract **knowledge and insights** from structured and unstructured data. It combines elements from: **Statistics, Computer Science, Mathematics, and Domain Expertise.**

Purpose of Data Science

1. **Discover Patterns and Trends** using historical data.
2. Strategic decisions using predictive models and statistical insights.
3. Build predictive models to forecast future trends.
4. Automate tasks using AI and machine learning.
5. **Improve Business Efficiency.**
6. **Support Innovation.**

Need of Data Science

1. **Companies generate Big Data.** Traditional tools cannot handle this scale and variety.
2. **Better Decision Making:** Allows organizations to **make informed decisions** based on evidence rather than intuition.
3. **Predictive Capabilities:** With ML, businesses can **predict outcomes** (customer churn, equipment failure, product demand).
4. **Personalization** in e-commerce, entertainment, education, etc.
5. **Automation and Efficiency:** Chatbots, automated fraud detection, predictive maintenance.
6. **Competitive Advantage** by understanding customer needs, optimizing operations, and innovating faster.

Data Science/Scientists process is **iterative** and **cyclical**, meaning steps can be revisited and improved over time.

1. **Problem Definition:** Understand what success looks like. Example: "Predict customer churn".
2. **Data Collection:** Gather data from various sources such as: Databases (SQL, NoSQL), APIs, Web scraping, Sensors, Excel files.
3. **Data Pre-processing**
4. **EDA (Exploratory Data Analysis):** Use statistical, visualization techniques to understand: Data distribution, Relationships, Patterns and trends.
5. **Feature Engineering:** Raw data in meaningful features, improves model performance. Techniques: Normalization/Standardization, One-hot encoding, Creating new variables
6. **Choose, apply ML algorithms: Supervised:** Linear Regression, Decision Tree, SVM, and **Unsupervised:** K-Means, PCA. Split data into training and testing sets. Train model using the training data.
7. **Model Evaluation** using metrics: Accuracy, Precision, Recall, F1-score, RMSE, MAE.
8. **Deployment** into a **production environment.** The model is now used to make real-time predictions.
9. **Continuously Monitoring Performance & Maintenance:** Retrain or update the model as new data becomes available. Address **data drift** and **concept drift** (when the data or its meaning changes over time).

Structured Data	Unstructured Data
Data organized in a predefined schema (rows and columns)	Data without a predefined format or model
Tabular format	Free-form format
Stored in relational databases	Data lakes, NoSQL, file systems
Easy to process using traditional tools like SQL	Requires advanced techniques like NLP, image/video processing
Employee records, transaction logs, sensor data	Emails, social media posts, customer reviews, multimedia files
SQL, Excel, BI tools	Hadoop, Spark, PyLib, cloud storage
smaller and well-defined	Large and complex
Reporting, dashboarding, business operations	Sentiment analysis, content recommendation

Quantitative Data	Categorical Data
Data that represents numeric values and quantities	Data that represents categories or groups
Numerical and measurable	Qualitative and descriptive
Discrete (countable) and Continuous (measurable)	Nominal (no order) and Ordinal (ordered categories)
Mathematical operations like addition, subtraction, etc.	Logical operations like grouping, counting, or frequency analysis
Age, income, height	Gender, country, product type
Histograms, box/scatter plots	Bar charts, pie charts
numeric data types (int, float)	String or label format

Directly used in regression, clustering, etc.	Often requires encoding (e.g., one-hot encoding)
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Small Data	Big Data
Data that is small in volume, manageable, and easily processed	Data that is large in volume, velocity, and variety (3Vs)
Fits in a single machine	From TB to PB or more
local systems, spreadsheets, traditional databases	distributed systems like Hadoop, cloud storage
Can be processed using a single computer and basic tools	Requires parallel and distributed computing frameworks
Excel, SQL, R, Python	Hadoop, Spark, cloud platforms
Sales record, survey data	Social media, sensor networks
Fast processing and analysis	computational time, resources
Simple and structured data	Complex any type of data

Feature	Business Intelligence	Data Science
Focus	Descriptive analytics – understanding historical data	Predictive and prescriptive analytics – forecasting and decision making
Objective	Reporting, monitoring KPIs, and answering "What happened?"	Extracting insights, building models to predict "What will happen?"
Data Types	Primarily structured data from internal sources	Structured, semi-structured, and unstructured data from diverse sources
Techniques	Querying, dashboards, OLAP cubes	Machine learning, statistical modeling, AI techniques
Tools	Power BI, Tableau, Excel, SAP BI	Python, R, TensorFlow, Hadoop, Spark
Output	Reports, dashboards, scorecards	Predictive models, recommendations, automated decisions
Users	Business analysts, managers	Data scientists, machine learning engineers
Time Frame	Past and present data analysis	Future forecasting and scenario simulation
Goal	Support business decision-making with historical insight	Automate and improve decision-making using data-driven predictions

Use of Data Science

1. **Business Optimization:** analyse customer behaviour, optimize supply chains, and improve marketing strategies.
2. **Healthcare:** assist in early diagnosis of diseases, personalized treatment plans, and managing hospital resources efficiently.
3. **Finance:** Fraud detection systems and risk assessment models.
4. **Manufacturing:** Predictive maintenance models anticipate equipment failures, reducing downtime and costs.
5. **Retail:** Recommendation engines personalize shopping experiences to increase customer engagement and sales.

Example: Customer Churn Prediction in Telecom Industry

Problem: Telecom companies want to predict which customers are likely to discontinue service (churn) to proactively retain them.

Data used: customer demographics, call detail records, billing information, and customer support interactions.

Data Science Process:

Outcome: The Company can target high-risk customers with retention offers, reducing churn rate and improving profitability.

Role	Responsibilities
Data Scientist	Develops ML models, performs statistical analysis, and derives insights from data.
Data Analyst	Examines datasets to summarize findings using statistical tools and visualizations.
Data Engineer	Designs, builds, and manages scalable data pipelines and architectures.
Machine Learning Engineer	Focuses on implementing and optimizing machine learning algorithms in production systems.
Business Analyst	Bridges the gap between business needs and data solutions by interpreting results and suggesting strategies.
Statistician	Applies mathematical theories and statistical methods to analyze and interpret complex data.
Data Architect	Designs, manages the overall data infrastructure, ensuring data accessibility and integrity.